

First, initiate N instances with a softmax distribution based on degree. Calculate instance fitness as the size of the clique. With a softmax-like tournament, perform the following:

Select N pairs of mates to breed. Cross over. The crossover process is like so: If the intersection is not null, take the intersection. If the intersection is null, see if the intersection between their candidate sets is not null. If the intersection between the candidate sets is null, choose a partner at random and just greedily add.

On a softmax random O nodes, perform k-opt.

Repeat.

Algorithm 1 Genetic Algorithm

1: **procedure** GENETICALGORITHM($G = (U, V, E), N, M$)
2:
3: **end procedure**
